

ESE 5410 Midterm 1 Study Guide

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Oct 4, 2026

Everything for Midterm 1 (Modules 2–6) in one place: notes, all 59 quiz questions solved, selected module practice problems, and the full midterm practice set, worked step by step.

How to use this guide

1. **Notes (Modules 2–6):** each concept follows the same pattern: a concrete example, the formula, a table explaining every symbol, and exam traps. Modules 2–3 are condensed because we went through them in chat.
2. **Quizzes, practice problems, midterm practice:** cover the solution, attempt the problem, then check. Doing the problems is where the points come from.
3. **Formula sheet** at the end: read it in the last hour before 6 PM.

Exam conditions to remember: Honorlock-proctored, so expect to compute by hand. Write your intermediate tables ($x - \bar{x}$, $y - \bar{y}$, products) neatly; most lost points are arithmetic slips.

Answer-key issues corrected in this guide:

- Midterm 8c: without using X , every classifier has AUC exactly $\frac{1}{2}$. You can do neither better nor worse.
- Midterm 7d: the key's reasoning is muddled. The noise mean gets absorbed into the intercept (explained there).
- Module 5 quiz (rating for a 50% win chance): the key's own work gives 1,625; the 1,600 shown is a rounding tolerance.
- Midterm Problem 3 defines $g_1(x) = \Pr(Y = 0 \mid x)$. That is *not* a typo; the solution is consistent with it (all 13 points checked).

Module 2: Statistical Learning

The one rule to own from this module: **expected squared error = (miss)² + Var(noise)**. Bias-variance and the irreducible-error questions all come from it.

2.1 Setup and notation

Symbol	Meaning
$\mathbf{X} = (X_1, \dots, X_p)$	Random vector of p inputs (e.g. TV, radio, newspaper spend: $p = 3$)
$\mathbf{x}$	One specific value of the inputs, e.g. (1000, 800, 900)
Y, y	The random output (sales) and a specific value of it
$D = \{(\mathbf{x}_i, y_i)\}, i = 1 \dots N$	The dataset. x_{ik} = feature k of sample i
f	The regression function: fixed, unknown , used by nature
$\hat{f}$	Your estimate of f (hat = estimated from data)

The course assumes the **additive model**:

$$Y = f(X) + \varepsilon, \quad E[\varepsilon] = 0, \quad \text{Var}(\varepsilon) = \sigma^2$$

Regression vs. classification: decided by the *output* only. A numeric output (price, test score, CO₂ level) is regression; a label (rise/fall, success tier, spam/not) is classification, even if the inputs are numeric or binary.

Prediction vs. inference: prediction = getting good $\hat{Y}$; inference = understanding *which* inputs affect Y and how (e.g. "which factors affect housing prices").

2.2 The regression function

$$f(\mathbf{x}) = E[Y | X = \mathbf{x}] = \int y f_{Y|X}(y | \mathbf{x}) dy$$

- **Meaning:** $f(x)$ is the average Y among all cases with input x.
- **Why:** fix $X = x$, so $Y = f(x) + \varepsilon$ and $E[Y | X = x] = f(x) + E[\varepsilon] = f(x)$.
- The integral is just the definition of an average (each y times its density). You never compute it on the exam. $f_{Y|X}$ is a density, unrelated to the regression function f.
- **If the noise mean is not zero** (e.g. $\varepsilon \sim N(1, 2)$), the mean is absorbed into f: $Y = X + X^2 + \varepsilon$ gives $f(x) = x + x^2 + 1$, irreducible error = $\text{Var}(\varepsilon) = 2$.

f is the best predictor under squared error:

$$f = \arg \min_g E[(Y - g(X))^2]$$

Piece	Meaning
g	Any candidate prediction function
$Y - g(X)$	The error (residual)
$(Y - g(X))^2$	Squared error: always positive, punishes big misses more
$E[\cdot]$	Average over all (X, Y): the mean squared error (MSE)
arg min over g	The g that achieves the smallest MSE (min = the value, arg min = the thing achieving it)

The key move (tested directly): fix $X = x$ and let $a = f(x) - g(x)$, a plain number.

$$E[(Y - g(x))^2 | X = x] = E[(a + \varepsilon)^2] = a^2 + 2a E[\varepsilon] + E[\varepsilon^2] = (f(x) - g(x))^2 + \sigma^2$$

- $(f(x) - g(x))^2$ = **reducible error**; zero only when $g = f$.
- σ^2 = **irreducible error**; even the perfect model pays it.
- **Recipe:** miss = $f(x) - g(x)$, answer = miss² + $\text{Var}(\varepsilon)$.

Facts you need cold:

- $E[\varepsilon^2] = \text{Var}(\varepsilon) + (E[\varepsilon])^2$. With mean 0, $E[\varepsilon^2] = \sigma^2$.
- **Uniform on [a, b]:** mean = $(a + b)/2$, **variance** = $(b - a)^2/12$. $\text{Unif}[-1, 1] \rightarrow 1/3$; $\text{Unif}[-2, 2] \rightarrow 4/3$.
- **Conditional distribution:** $(Y | X = x) = f(x) + \varepsilon$ is the noise distribution shifted by $f(x)$. Example: $Y = X^2 + \varepsilon$, $\varepsilon \sim \text{Unif}[-1, 1] \rightarrow (Y | X = 1/2) \sim \text{Unif}[-3/4, 5/4]$; $\text{Var}(Y | X = x) = \text{Var}(\varepsilon) = 1/3$.

2.3 Estimating f by local averaging

Exact matches $x_i = x$ essentially never occur with continuous inputs, so average over a window of radius r:

$$D_r(\mathbf{x}) = \{i: \|\mathbf{x}_i - \mathbf{x}\| \leq r\}, \quad \hat{f}(\mathbf{x}) = \frac{1}{|D_r(\mathbf{x})|} \sum_{i \in D_r(\mathbf{x})} y_i$$

Symbol	Meaning
$\{i : \dots\}$	The set of data points i satisfying the condition
$\ x_i - x\ $	Distance between point i and the target x
$ D_r(x) $	How many points are in the window
$\hat{f}(x)$	Average of their y values

Sliding the window along traces a curve through the middle of the cloud (essentially KNN).

2.4 The curse of dimensionality

Idea: every extra feature is another “similar to mine” filter, and filters multiply. In high dimensions a small window contains almost no data, so local averaging and KNN fail.

Density × volume. N points spread uniformly over region S :

$$E[\text{\# points in window}] = \frac{N}{\text{Vol}(S)} \times \text{Vol}(\text{window})$$

p	Region	Window	Expected count
1	$[-1, 1]$, length 2	length $2r$	Nr
2	Unit disc, area π	area πr^2	Nr^2
p	Unit hypersphere, volume K_p	volume $K_p r^p$	$N r^p$

K_p cancels; only “volume scales as r^p ” matters. Lecture simulation: $N = 1000$, $r = 0.1$ gives 100, 10, 1, ..., 0.01 points at $p = 1, 2, 3, \dots, 5$.

Hypercube version (Module 2 quiz). Point a uniform in $H_1 = [-1, 1]^p$; inner cube $H_r = [-r, r]^p$:

$$P(a \in H_r) = \frac{(2r)^p}{2^p} = r^p, \quad P(a \notin H_r) = 1 - r^p$$

- 90% outside $\rightarrow r^p = 0.1 \rightarrow r = 10^{-(1/p)}$. As $p \rightarrow \infty$, $r \rightarrow 1$: almost all volume sits in a **thin shell near the surface**.
- Sphere inside cube** (Module 2 practice): the fraction of the cube $[-1, 1]^p$ inside the inscribed unit sphere is about 0.785 at $p = 2$, 0.52 at $p = 3$, and essentially 0 by $p \approx 10$.

Distances grow with p. a, b independent uniform on $[-1, 1]^p$ (mean 0, $E[a_i^2] = \text{Var} = 1/3$):

$$E\|a - b\|^2 = \sum_{i=1}^p E[(a_i - b_i)^2] = p(E[a_1^2] + E[b_1^2] - 2E[a_1]E[b_1]) = \frac{2p}{3}$$

To the origin ($b = 0$): $E\|a\|^2 = p/3$. Every point is far from every other point, so there are no real neighbours.

2.5 Parametric models

Assume a shape with parameters θ and learn θ from **all** the data, which is why parametric models beat the curse:

$$f_L(x; \beta) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p \quad f(x; \alpha) = \alpha_0 + \alpha_1 x + \dots + \alpha_d x^d$$

Here $x_1 \dots x_p$ are the features of one input, not data points. Polynomial degree d = flexibility: in the lecture, $d = 2$ is OK, $d = 3$ is best, $d = 20$ wiggles through the noise (**overfitting**). Other parametric models named: regression trees, SVMs. The catch: a wrong assumed shape costs bias.

2.6 Train vs. test MSE

$$\text{MSE}_{Tr} = \frac{1}{N} \sum_{i \in D_{Tr}} (y_i - \hat{f}(\mathbf{x}_i))^2 \quad \text{MSE}_{Te} = \frac{1}{M} \sum_{i \in D_{Te}} (y_i - \hat{f}(\mathbf{x}_i))^2$$

As flexibility increases	Behaviour	Why
Training MSE	Always decreases (or stays flat)	More flexible models bend closer to the training points
Test MSE	U-shaped	First captures the real pattern, then starts fitting noise

Left of the minimum = underfitting, right = overfitting. **Choose flexibility by minimizing test MSE, never training MSE.**

2.7 Bias-variance tradeoff

Imagine retraining on many independent training sets and predicting at a test point $\mathbf{x}_0$:

$$E[(y_0 - \hat{f}(x_0))^2] = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\varepsilon)$$

$$\text{Bias}(\hat{f}(x_0)) = E[\hat{f}(x_0)] - f(x_0) \quad \text{Var}(\hat{f}(x_0)) = E[(\hat{f}(x_0) - E[\hat{f}(x_0)])^2]$$

- Bias = how far the **average** prediction is from the truth (systematic miss).
- Variance = how much the prediction **jumps** between training sets.
- Bias² + Variance = reducible; Var(ε) = irreducible.
- **Example:** truth $f(x_0) = 10$. Rigid model predicts 7, 7, 8, 8 → Bias² = 6.25, Var = 0.25. Flexible model predicts 6, 14, 9, 11 → Bias = 0, Var = 8.5.

Curve vs. flexibility	Shape
Squared bias	Starts high, decreases
Variance	Starts low, increases
Irreducible error Var(ε) ("Bayes error")	Flat dashed line
Test MSE = sum of the three	U-shaped, always above the dashed line
Training MSE	Decreasing, can go below the dashed line

Degree (lecture)	10 fits on 10 datasets	Bias	Variance
d = 1	Nearly identical lines	High	Low (underfit)
d = 3	All track the middle of the cloud	Low	Low
d = 20	Wiggly, all different	Low	High (overfit)

When does a flexible method win? (Module 2 practice)

Situation	Flexible vs. inflexible	Reason
N huge, p small	Flexible better	Enough data to keep variance low
p huge, N small	Flexible worse	Overfits the few points
Truth highly nonlinear	Flexible better	Inflexible model has high bias
Noise variance σ^2 very high	Flexible worse	Fits the noise

Traps: overfitting comes from high *variance*, not high bias. Test MSE can never be below $\text{Var}(\epsilon)$. Training MSE never goes up when flexibility increases (nested models).

Module 3: Linear Regression

The most exam-heavy module: one recipe (fit $\rightarrow$ RSS/TSS/ R^2 /RSE $\rightarrow$ SDs $\rightarrow$ confidence intervals $\rightarrow$ tests) that appears in midterm Problems 2 and 7.

3.1 Linear algebra review

Rule	Detail
Matrix A is $m \times n$	m rows, n columns; A_{ij} = row i , column j
Transpose A^T	Rows become columns; $(A^T)^T = A$; $(A + B)^T = A^T + B^T$; $(AB)^T = B^T A^T$ (order flips)
Product $(m \times n)(n \times p) = (m \times p)$	Inner sizes must match; $C_{ij} = \sum_k A_{ik} B_{kj}$ (row i of $A \cdot$ column j of B)
Properties	Associative $(AB)C = A(BC)$; distributive $A(B + C) = AB + AC$; not commutative : $AB \neq BA$ in general
Identity I_n	1s on the diagonal, 0s elsewhere; $AI = IA = A$
Vector $x \in \mathbb{R}^n$	A column ($n \times 1$) by convention; entries x_i
Inner product	$x^T y = \sum x_i y_i$, a scalar; $x^T y = y^T x$
Outer product	xy^T , an $n \times n$ matrix with entry $(i, j) = x_i y_j$
Matrix $\times$ vector	$(m \times n)(n \times 1) = m$ -vector; $y_i = \sum_j A_{ij} x_j$
Eigenpair	$Mv = \lambda v$ (square M). Symmetric M ($M = M^T$): n real eigenvalues, real eigenvectors, mutually perpendicular
2×2 inverse	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = 1/(ad - bc) \cdot \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

3.2 The model and least squares

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon$$

The true β 's are deterministic but unknown (only nature knows them); $\hat{\beta}$ with hats are your estimates. Choose $\hat{\beta}$ to minimize the **residual sum of squares**:

$$\text{RSS} = \sum_{i=1}^N \left(y_i - \hat{y}_i \right)^2 = \| \mathbf{y} - M_X \theta \|^2, \quad \hat{\beta} = \left(M_X^T M_X \right)^{-1} M_X^T \mathbf{y}$$

Symbol	Meaning
$y_i - \hat{y}_i$	Residual: vertical gap between point and line
M_X	Data matrix: one row per point, first column all 1s , then the features
$\mathbf{y}$	Column of all outputs
$(M_X^T M_X)^{-1} M_X^T$	The pseudo-inverse

One input ($p = 1$): use these on the exam.

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

- $\hat{\beta}_0$ = intercept, $\hat{\beta}_1$ = slope, $\bar{x}$ and $\bar{y}$ = sample means.
- **The fitted line always passes through $(\bar{x}, \bar{y})$:** $\hat{y}(\bar{x}) = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 \bar{x} = \bar{y}$.
- If $\bar{x} = 0$, the formulas simplify to $\hat{\beta}_1 = \sum x_i y_i / \sum x_i^2$ and $\hat{\beta}_0 = \bar{y}$.

Worked example. Data (1, 2), (2, 3), (3, 5), (4, 6), (5, 9); $\bar{x} = 3$, $\bar{y} = 5$.

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	product	$(x_i - \bar{x})^2$	$\hat{y}_i$	residual ²	$(y_i - \bar{y})^2$
1	2	-2	-3	6	4	1.6	0.16	9
2	3	-1	-2	2	1	3.3	0.09	4
3	5	0	0	0	0	5.0	0	0
4	6	1	1	1	1	6.7	0.49	1
5	9	2	4	8	4	8.4	0.36	16
Sum				17	10		RSS = 1.1	TSS = 30

$\hat{\beta}_1 = 17/10 = 1.7$; $\hat{\beta}_0 = 5 - 1.7 \times 3 = -0.1$; line $\hat{y} = -0.1 + 1.7x$.

3.3 Model accuracy: RSS, TSS, R^2 , RSE

Metric	Formula	Meaning
RSS	$\sum (y_i - \hat{y}_i)^2$	Error left with the line
TSS	$\sum (y_i - \bar{y})^2$	Error predicting $\bar{y}$ every time (= RSS of the model with $\beta_1 = 0$, no X)
R^2	$(TSS - RSS)/TSS = 1 - RSS/TSS$	Fraction of squared error removed by using X; between 0 and 1
RSE	$\sqrt{RSS/(N - p - 1)}$	Estimate of σ (noise SD); $p = 1 \rightarrow$ divide by $N - 2$

Example: $R^2 = 1 - 1.1/30 = 0.963$; $RSE = \sqrt{(1.1/3)} = 0.606$.

3.4 Coefficient uncertainty

Data are random, so $\hat{\beta}$ is a random variable. Facts:

- **Unbiased:** $E[\hat{\beta}_i] = \beta_i$.
- **Covariance matrix** (given the inputs): $\text{Cov}(\hat{\beta}) = \sigma^2(M_X^T M_X)^{-1}$. Its diagonal entries are the variances of the $\hat{\beta}_i$.
- **Approximately normal.** Lecture demo: 1,000 datasets of $N = 100$ with $\beta_1 = 0.5$ gave a bell curve centred at 0.5 with SD ≈ 0.05 .

$$SD(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum (x_i - \bar{x})^2} \quad SD(\hat{\beta}_0)^2 = \sigma^2 \left(\frac{1}{N} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right)$$

Symbol	Meaning
$SD(\hat{\beta})$	Also written SE (standard error): how much $\hat{\beta}$ would jump between datasets
σ^2	Variance of the measurement noise ε (a common quiz question)
$\Sigma(x_i - \bar{x})^2$	Spread of the inputs: spread-out x's give a more precise slope

Which σ^2 to use: the noise variance if the problem gives it ($N(0, 8s^2)$, $N(0, w^2)$); otherwise **RSE²**.

Don't swap them (your quiz mistake): slope = $\sigma^2 \div x\text{-spread}$ (simple). Intercept = the one with the bracket ($1/N + \bar{x}^2/\dots$); when $\bar{x} = 0$ it is σ^2/N , the variance of an average, because then $\hat{\beta}_0 = \bar{y}$.

Example (RSE² = 0.367): $SD(\hat{\beta}_1) = \sqrt{(0.367/10)} = 0.19$; $SD(\hat{\beta}_0) = \sqrt{(0.367 \times (0.2 + 0.9))} = 0.64$.

3.5 Confidence intervals

95% of a normal distribution lies within ± 2 SDs of its mean, so:

$$P(\beta_1 \in [\hat{\beta}_1 - 2SD(\hat{\beta}_1), \hat{\beta}_1 + 2SD(\hat{\beta}_1)]) = 0.95$$

Start from $\hat{\beta}_1 \in [\beta_1 \pm 2SD]$ w.p. 95% and rearrange; the hat moves to the interval's centre. Same for β_0 . Example: $\beta_1 \in [1.32, 2.08]$, $\beta_0 \in [-1.37, 1.17]$.

Warnings (lecture): this assumes $\hat{\beta}$ is approximately normal (not always true) and that σ^2 is known or well estimated.

3.6 Hypothesis testing

- H₀:** $\beta_1 = 0$ (X does not influence Y). **H_a:** $\beta_1 \neq 0$.
- Checking whether $\hat{\beta}_1 = 0$ exactly is useless: a continuous random variable equals an exact value with probability 0 (like a person weighing exactly 82.371817 kg).

$$t_1 = \frac{\hat{\beta}_1 - 0}{SD(\hat{\beta}_1)} \sim N(0, 1) \text{ under } H_0$$

- |t| > 2 $\rightarrow$ reject H₀** (statistically significant). $|t| \leq 2 \rightarrow$ fail to reject.
- Equivalent:** reject H₀: $\beta = c$ iff c lies outside the 95% CI. To test $c \neq 0$, use the CI or $t = (\hat{\beta} - c)/SD$.
- Under H₀ a value outside ± 2 happens only 5% of the time: rare, not impossible.
- Judge significance by t, never by the coefficient's size.** 50 with SD 40 is insignificant; 0.046 with SD 0.0014 is hugely significant.
- Example: $t_1 = 1.7/0.19 = 8.9 \rightarrow$ reject; $t_0 = -0.1/0.64 = -0.16 \rightarrow$ fail to reject.

Legal parable (H₀ = innocent until proven guilty):

Outcome	Legal analogy	What you may say
Reject H ₀	Convicted	Strong statistical evidence, but H ₀ could still be true
Fail to reject H ₀	Not guilty	Not proof of innocence; never "accept" or "prove" H ₀ . In practice you act as if H ₀ holds (e.g. drop the variable)

3.7 "Solve for s" problems

When the noise is $N(0, s^2)$ with s unknown, the SDs and CIs depend on s . Example: same data, $H_0: \beta_1 = 1$.

1. $SD(\hat{\beta}_1) = \sqrt{s^2/10} = s/\sqrt{10}$; $CI = 1.7 \pm 2s/\sqrt{10}$.
2. Reject $\leftrightarrow$ 1 outside the CI. Since $1 < 1.7$: $1 < 1.7 - 2s/\sqrt{10}$.
3. $s < 0.35\sqrt{10} \approx 1.11$.

Less noise $\rightarrow$ narrower CI $\rightarrow$ easier to reject. "Reject" questions give $s < \dots$; "cannot reject" give $s > \dots$

3.8 Exam recipe ($p = 1$)

1. $\bar{x}, \bar{y}$.
2. Table: $x - \bar{x}, y - \bar{y}$, product, $(x - \bar{x})^2$.
3. $\hat{\beta}_1 = \Sigma \text{product} / \Sigma(x - \bar{x})^2$; $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$.
4. $\hat{y}_i$, residuals $\rightarrow$ RSS; TSS; $R^2 = 1 - \text{RSS}/\text{TSS}$; $\text{RSE} = \sqrt{(\text{RSS}/(N - 2))}$.
5. $\sigma^2 =$ given noise variance, else RSE^2 .
6. $SD(\hat{\beta}_1)$, $SD(\hat{\beta}_0)$ from the formulas above.
7. $CI = \hat{\beta} \pm 2SD$; reject if $|t| > 2$ or the null value is outside the CI.

Module 4: Extended Linear Regression

Categorical inputs, polynomials, and interactions are all ordinary linear regression with **made-up input columns**; the exam skill is splitting the model into one equation per case and reading the coefficients.

4.1 What linear regression accepts

Inputs can be **quantitative or qualitative** (coded as dummies); the output must be **quantitative**. A coded category is still a qualitative variable. (Module 4 quiz.)

4.2 Categorical input with two levels

Example: credit-card balance vs. gender, nothing else.

1. Dummy: $x_i = 1$ if person i is female, 0 if male.
2. Fit $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, which splits into two cases:

$$y_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{female } (x_i = 1) \\ \beta_0 + \varepsilon_i & \text{male } (x_i = 0) \end{cases}$$

Coefficient	Meaning
β_0	Average balance of the baseline group (male)
β_1	Difference: female average minus male average

Test $H_0: \beta_1 = 0$ (gender doesn't affect balance) with $t = \hat{\beta}_1/SD(\hat{\beta}_1)$. In the lecture's table $|t| < 2$, so we **cannot reject**: no strong evidence gender matters (not proof that it doesn't).

4.3 Categorical input with K levels

Example: ethnicity $\in \{\text{Caucasian, African American, Asian}\}$.

1. Pick a **baseline** (lecture: African American). Any level works.

2. Create **K - 1 = 2 dummies**: $x_{i1} = 1$ if Asian; $x_{i2} = 1$ if Caucasian.

3. Fit $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$:

Group	x_{i1}	x_{i2}	Mean balance
Asian	1	0	$\beta_0 + \beta_1$
Caucasian	0	1	$\beta_0 + \beta_2$
African American (baseline)	0	0	β_0

β_1 and β_2 are differences **from the baseline**. Testing $\beta_1 = 0$ asks "does being Asian (vs. baseline) change balance?" In the lecture both $|t| \leq 2 \rightarrow$ cannot reject either.

Why K - 1 and not K: the baseline is already captured by β_0 ; a K-th dummy would duplicate the intercept.

4.4 Nonlinear effects: polynomial regression

Example: 392 cars, mpg vs. horsepower. Treat powers as new inputs:

$$\text{mpg} = \beta_0 + \beta_1 \text{HP} + \beta_2 \text{HP}^2 + \dots + \beta_k \text{HP}^k + \varepsilon, \quad \mathbf{x}_i = (\text{HP}_i, \text{HP}_i^2, \dots, \text{HP}_i^k)$$

- It's still **linear in the β 's**, so the same $\hat{\beta} = (\mathbf{M}_X^T \mathbf{M}_X)^{-1} \mathbf{M}_X^T \mathbf{y}$ applies. The model is nonlinear in X, linear in the parameters.
- $\mathbf{M}_X$ gets columns $[1, x, x^2, \dots]$. For the Module 4 quiz data (1, 2), (2, 5), (3, 13), (4, 20): $\mathbf{M}_X$ rows are $[1, x_i, x_i^2]$; $\mathbf{M}_X^T$ is 3×4 with rows (1, 1, 1, 1), (1, 2, 3, 4), (1, 4, 9, 16).
- Lecture: degree 2 fit best; all three t-statistics > 2 , so HP^2 is a relevant input.
- **Substitution trick** (Module 3-4 practice P6): for $\hat{Y} = \alpha X^2 + \gamma$, set $Z = X^2$ and run simple regression of Y on Z.

4.5 Interaction terms (quantitative $\times$ quantitative)

The plain model assumes TV and radio act independently. If radio boosts TV's effect (synergy), add the product as a third input:

$$\text{sales} = \beta_0 + \beta_1 \text{TV} + \beta_2 \text{radio} + \beta_3 (\text{TV} \times \text{radio}) + \varepsilon$$

Fit by treating $\text{TV} \times \text{radio}$ as an ordinary feature ($p = 3$). Lecture: all $|t| > 2$, so the interaction matters.

Hierarchy principle: if you include an interaction $X_1 X_2$, **also keep the main terms X_1 and X_2** , even if their t-statistics are below 2. The interaction is hard to interpret without them.

4.6 Interaction with a categorical input

Example: balance vs. income (number) and student (dummy).

Without interaction: $\text{balance} = \beta_0 + \beta_1 \cdot \text{income} + \beta_2 \cdot \text{student}$.

Group	Intercept	Slope
Student	$\beta_0 + \beta_2$	β_1
Non-student	β_0	β_1

Two **parallel** lines: only the intercept shifts.

With interaction: $\text{balance} = \beta_0 + \beta_1 \cdot \text{income} + \beta_2 \cdot \text{student} + \beta_3 \cdot (\text{income} \times \text{student})$.

Group	Intercept	Slope
Student	$\beta_0 + \beta_2$	$\beta_1 + \beta_3$
Non-student	β_0	β_1

The lines have **different intercepts and different slopes**, so they can cross.

Worked example (midterm P6 / Module 4 quiz). $\hat{Y} = 35 + 5 \cdot \text{GPA} - 15 \cdot \text{Gender} + 5 \cdot (\text{GPA} \times \text{Gender})$, Gender = 1 for female.

1. Male (Gender = 0): $\hat{Y} = 35 + 5 \cdot \text{GPA}$.
2. Female (Gender = 1): $\hat{Y} = (35 - 15) + (5 + 5) \cdot \text{GPA} = 20 + 10 \cdot \text{GPA}$.
3. Crossing: $35 + 5g = 20 + 10g \rightarrow g = 3.0$. Females earn more for $\text{GPA} > 3.0$; males for $\text{GPA} < 3.0$.

Traps:

- “ β_3^* is large, so the interaction is significant” is **false**: check its t-statistic.
- Adding a term (e.g. $\text{GPA}^2 \times \text{Gender}$) never raises **training** RSS; whether **test** RSS goes down is unknown (may overfit).
- If an added term is multiplied by Gender, it vanishes for males: on male-only data both models give identical predictions and identical training RSS.

Module 5: Classification and Logistic Regression

Two skills carry this module: **find the Bayes classifier** by solving $p_1(x) \geq \frac{1}{2}$ (or taking the highest curve), and **compute with the logistic function** via log-odds.

5.1 The classification setup

- Output Y is **qualitative**: it takes one of K labels in a set $\mathcal{C}$ (e.g. {dog, cat}).
- A **classifier** $C(x)$ maps an input vector $x \in \mathbb{R}^p$ to a label in $\mathcal{C}$. It will get some inputs wrong; those errors are studied in Module 6.
- Classification vs. regression is decided by the **output** type only (Module 5 quiz: “rise or fall” and “three success tiers” are classification; a test score or a CO₂ amount is regression).

5.2 The generative model

Nature makes each data point in two steps:

1. Draw an input x_i from the marginal f_X .
2. Draw its label from the **conditional class probability**:

$$p_k(\mathbf{x}) = \Pr(Y = k \mid X = \mathbf{x}), \quad \sum_k p_k(\mathbf{x}) = 1$$

Example: $X \sim \text{Unif}[-2, 2]$, p_1 a sigmoid. Draw $x_1 = 0.44 \rightarrow p_1 = 0.70$, $p_0 = 0.30 \rightarrow$ **flip a coin with those biases** (not “pick the larger”) $\rightarrow$ say $y_1 = 1$. Draw $x_2 = -1.1 \rightarrow p_1 \approx 0.1 \rightarrow$ most likely $y_2 = 0$. Repeat N times. Where p_1 is near 1, almost all labels are 1, with a few 0s.

5.3 The Bayes optimal classifier

$$C(\mathbf{x}) = \arg \max_k p_k(\mathbf{x})$$

- For each x , output the label with the **highest conditional class probability**.

- It **minimizes the total classification error rate** (best when false positives and false negatives cost the same). It does *not* minimize false negatives alone: "always predict 1" has zero FNs.
- **Binary shortcut:** $p_0 = 1 - p_1$, so $p_1 \geq p_0 \Leftrightarrow \mathbf{p_1(x) \geq 1/2}$. Predict 1 there, 0 elsewhere (ties may go either way).

Worked examples:

$p_1(x)$	Solve $p_1 \geq 1/2$	Bayes classifier
Sigmoid centred at 0	$x \geq 0$	1 if $x \geq 0$, else 0
$1/(1 + x^2)$	$2 \geq 1 + x^2 \rightarrow x^2 \leq 1$	1 if $x \in [-1, 1]$, else 0
$(x^2 - 1/2)/(x^2 + 1/2)$	$2x^2 - 1 \geq x^2 + 1/2 \rightarrow x^2 \geq 3/2$	1 if $ x \geq \sqrt{6}/2 \approx 1.22$, else 0

More than two classes: sketch every $p_k(x)$ on one axis; the classifier follows the **top curve**, and boundaries are where the top curve changes. Solve each crossing as an equation between the two curves (see the triangular-function quiz).

Function shifts (used to sketch these): $R(x + 1)$ is $R(x)$ shifted **left** by 1; $R(x - 1)$ shifted right.

5.4 Estimating p_k by local averaging (and why it fails)

Nature's p_k is unknown. Local averaging: take a window around x (e.g. $x = -0.2$, window $[-0.4, 0]$) and set

$$\widehat{p}_1(x) = \frac{\#\{\text{label-1 points in the window}\}}{\#\{\text{points in the window}\}}, \quad \widehat{C}(x) = 1 \text{ if } \widehat{p}_1(x) \geq \widehat{p}_0(x)$$

It approximates the true curve in 1-D (the estimated threshold lands near, not exactly at, 0). In high dimensions the **curse of dimensionality** empties the windows, so we again switch to a **parametric** model.

5.5 The logistic (sigmoid) model

Binary classes $\{0, 1\}$, one input:

$$p_1(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}, \quad p_0(x) = \frac{1}{1 + e^{\beta_0 + \beta_1 x}} = 1 - p_1(x)$$

Parameter	Effect on the S-curve
β_1	Steepness: bigger $ \beta_1 $ = sharper jump; $\beta_1 = 1/2$ is flatter than 1, 2 is steeper; negative β_1 flips the direction
β_0	Shift left/right; the curve crosses $1/2$ where $\beta_0 + \beta_1 x = 0$, i.e. $x = -\beta_0/\beta_1$

With $\beta_0 = 0$, $\beta_1 = 1$: $p_1(0) = 1/2$, $p_1 \rightarrow 1$ as $x \rightarrow \infty$, $p_1 \rightarrow 0$ as $x \rightarrow -\infty$.

Sigmoid identities (Module 5–6 practice P2), with $S(x) = 1/(1 + e^{-x})$:

- $S(x) = e^x/(e^x + 1)$ (multiply top and bottom by e^x).
- $S(-x) = 1/(1 + e^x) = e^{-x}/(e^{-x} + 1)$.
- $1 - S(x) = S(-x)$. This is why $p_0 = 1 - p_1$ has the mirrored S shape.

5.6 Odds and log-odds

$$\text{odds} = \frac{p(x)}{1 - p(x)} \in [0, \infty), \quad \ln \frac{p(x)}{1 - p(x)} = \beta_0 + \beta_1 x$$

- Odds rise slowly, then blow up as $p \rightarrow 1$ (quiz: Plot B). $p = 1/2 \Leftrightarrow \text{odds} = 1 \Leftrightarrow \text{log-odds} = 0$.
- **The log-odds (logit) is linear in x .** That is the whole model.

- **Derivation** (the interview question): odds = $e^{\{\beta_0 + \beta_1 x\}} \rightarrow p = (1 - p)e^{\{\beta_0 + \beta_1 x\}} \rightarrow p = e^{\{\dots\}} - p \cdot e^{\{\dots\}} \rightarrow p(1 + e^{\{\dots\}}) = e^{\{\dots\}} \rightarrow p = e^{\{\dots\}} / (1 + e^{\{\dots\}})$.
- **Interpretation**: +1 unit of x adds β_1 to the **log-odds** (multiplies the odds by $e^{\{\beta_1\}}$). It does **not** add β_1 to the probability.

Prediction recipe: $z = \beta_0 + \beta_1 x_1 + \dots$; $p = e^z / (1 + e^z) = 1 / (1 + e^{-z})$. **For a 50% chance**: set $z = 0$ and solve for the unknown input.

Example: $\beta_0 = -6$, $\beta_1 = 0.05$ (hours), $\beta_2 = 1$ (GPA). 40 h, GPA 3.5: $z = -6 + 2 + 3.5 = -0.5 \rightarrow p = 0.378$. For 50%: $-6 + 0.05h + 3.5 = 0 \rightarrow h = 50$ hours.

5.7 Fitting: maximum likelihood

Assume the inputs are given. The **likelihood** is the probability of the observed labels as a function of candidate parameters θ ; independence makes it a product:

$$L(\theta_0, \theta_1) = \prod_{i: y_i = 1} p_1(x_i; \theta) \prod_{i: y_i = 0} p_0(x_i; \theta), \quad (\hat{\beta}_0, \hat{\beta}_1) = \arg \max_{\theta} L(\theta)$$

- Points labelled 1 contribute p_1 , points labelled 0 contribute p_0 .
- x_i and y_i are known, so L depends only on θ .

Coin example (midterm P5): H, H, T $\rightarrow L(p) = p^2(1 - p)$. 20 heads and 10 tails in any order $\rightarrow C(30, 20) \cdot p^{20}(1 - p)^{10}$. To maximize, take logs and differentiate: $2/p - 1/(1 - p) = 0 \rightarrow \hat{p} = 2/3$ (the observed fraction of heads), and the same 2/3 for 20/30.

Default dataset (lecture): balance predicts default; income barely does. With balance only: \$1,000 $\rightarrow$ 0.6% default probability; \$2,000 $\rightarrow$ about 58%.

5.8 Is an input relevant? The z-test

$\hat{\beta}_1$ from maximum likelihood is a random variable, approximately normal and unbiased; software gives its SD.

$$z_1 = \frac{\hat{\beta}_1}{SD(\hat{\beta}_1)}, \quad |z_1| > 2 \Rightarrow \text{reject } H_0: \beta_1 = 0$$

The z-statistic is logistic regression's t-statistic (not the z-score quantile). Default example: balance $|z| > 2 \rightarrow$ relevant.

5.9 Multiple inputs and multiple classes

$$p_1(\mathbf{x}) = \frac{e^{\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p}}{1 + e^{\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p}}$$

- Categorical inputs enter as dummies (student = 1/0), exactly as in Module 4.
- Default with balance, income, student: income has $|z| < 2 \rightarrow$ not relevant. **After dropping a variable, refit**; the old coefficients are no longer valid.

Multiclass logistic (K classes): each class k has its own coefficients $\beta_{0k}, \dots, \beta_{pk}$ (first index = which input, second = which class):

$$\Pr(Y = k | \mathbf{x}) = \frac{e^{\beta_{0k} + \beta_{1k} x_1 + \dots + \beta_{pk} x_p}}{\sum_{l=1}^K e^{\beta_{0l} + \beta_{1l} x_1 + \dots + \beta_{pl} x_p}}$$

Fitted by maximum likelihood the same way; then plug in $\hat{\beta}_k$ and classify by the largest.

Module 6: LDA and Classification Errors

LDA classifies by the larger of $\pi_k f_k(x)$; with two equal-variance Gaussians the boundary is one threshold γ , and every error rate is an area under a bell curve on the wrong side of γ .

6.1 LDA's generative model (the reverse of logistic)

	Logistic regression	LDA
Step 1	Draw input x from f_X	Draw label k with prior $\pi_k = \Pr(Y = k)$, $\sum \pi_k = 1$
Step 2	Draw label from $p_k(x)$	Draw input from the class density $f_k(x) = \Pr(X = x \mid Y = k) = N(\mu_k, \Sigma_k)$
Models directly	$\Pr(Y \mid X)$	$\Pr(X \mid Y)$

f_k is **not** the conditional class probability; the conditioning is reversed. Bayes' theorem turns it around:

$$p_k(\mathbf{x}) = \Pr(Y = k \mid X = \mathbf{x}) = \frac{\pi_k f_k(\mathbf{x})}{\sum_{l=1}^K \pi_l f_l(\mathbf{x})}$$

6.2 The LDA classifier

The denominator is the same for every k , so the largest p_k belongs to the largest numerator (5, 7, 3 and 5/15, 7/15, 3/15 peak at the same index):

$$C(\mathbf{x}) = \arg \max_k \pi_k f_k(\mathbf{x})$$

Recipe: draw each $\pi_k f_k(x)$ (a bell scaled by its prior); predict the class whose scaled bell is on top. The boundary γ is where two scaled bells cross.

How priors move the threshold ($\mu_1 = -1.5$, $\mu_2 = 1.5$, $\sigma^2 = 1$):

Priors	Crossing γ	Classifier
$\pi_1 = \pi_2 = 0.5$	0 (midpoint of the means)	Class 1 if $x < 0$, class 2 if $x \geq 0$
$\pi_1 = 0.3$, $\pi_2 = 0.7$	≈ -0.28 (shifts toward the rarer class)	Class 1 if $x < \gamma$, class 2 if $x \geq \gamma$

Two classes, one input, equal variance: the threshold formula (derived by taking logs of $\pi_1 f_1 = \pi_2 f_2$):

$$\gamma = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_2 - \mu_1} \ln \frac{\pi_1}{\pi_2}$$

Equal priors $\rightarrow \ln 1 = 0 \rightarrow \gamma = \text{midpoint}$. Check: $\ln(0.3/0.7)/3 = -0.847/3 = -0.28$.

6.3 The linear discriminant (univariate, shared variance)

log is increasing, so maximizing $\pi_k f_k =$ maximizing its log. Dropping terms that are the same for every class leaves

$$\delta_k(x) = x \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \ln \pi_k, \quad C(x) = \arg \max_k \delta_k(x)$$

δ_k is **linear in x** (a constant plus a coefficient times x), hence "linear" discriminant analysis.

6.4 Estimating the parameters from data

$$\hat{\pi}_k = \frac{N_k}{N}, \quad \hat{\mu}_k = \frac{1}{N_k} \sum_{i: y_i = k} x_i, \quad \hat{\sigma}^2 = \frac{1}{N-K} \sum_{k=1}^K \sum_{i: y_i = k} (x_i - \hat{\mu}_k)^2$$

Symbol	Meaning
N_k	Number of training points in class k
$\hat{\mu}_k$	Average input of class k
$\hat{\sigma}^2$	Pooled variance: squared distance of each point from its own class mean, divided by N – K

Plugging in estimates gives a threshold near, not exactly at, nature's (lecture: slightly below 0).

Several inputs (p > 1): $\mathbf{x}_k$ is a multivariate normal $N(\mu_k, \Sigma)$ with a **shared** p × p covariance Σ ; μ_k are points in the plane, level sets are ellipses, and the boundaries between classes are straight lines.

6.5 The four outcomes

	Actual 1	Actual 0
Predicted 1	True positive (TP)	False positive (FP)
Predicted 0	False negative (FN)	True negative (TN)

Cancer biopsy: FP = told "cancer" but healthy (more tests follow); FN = told "healthy" but has cancer (much worse). Counting formally, with $\delta(z) = 1$ only when $z = 0$:

$$FP = \sum_{i=1}^N \delta(C(\mathbf{x}_i) - 1) \delta(y_i), \quad FN = \sum_{i=1}^N \delta(C(\mathbf{x}_i)) \delta(y_i - 1), \quad \text{error rate} = \frac{FP + FN}{N}$$

6.6 Confusion matrix and the rates

Default dataset with LDA (N = 10,000; 9,667 paid, 333 defaulted):

	Actual 0 (paid)	Actual 1 (defaulted)	Row total
Predicted 0	TN = 9,644	FN = 252	9,896
Predicted 1	FP = 23	TP = 81	104
Column total	9,667	333	10,000

- Error rate = (23 + 252)/10,000 = **2.75%**.
- **Null classifier** (predict 0 for everyone): error = 333/10,000 = **3.33%**. LDA is barely better.
- Per class: 23/9,667 ≈ 0.2% of payers misclassified, but 252/333 ≈ **75%** of defaulters missed. The overall error rate hides this.

The four rates (divide by the **actual** column total):

Rate	Formula	Meaning	Default
TPR (sensitivity)	$TP/(TP + FN)$	Share of actual 1s caught	24%
FNR	$FN/(TP + FN) = 1 - TPR$	Share of actual 1s missed	76%
TNR	$TN/(TN + FP)$	Share of actual 0s correctly cleared	99.8%
FPR	$FP/(TN + FP) = 1 - TNR$	Share of actual 0s wrongly flagged	0.2%

Spam version: FPR = share of real emails sent to spam; FNR = share of spam let into the inbox; TPR = share of spam caught.

6.7 Expected errors in LDA = areas under curves

Two classes, one input, threshold γ (predict 1 for $x \geq \gamma$). By the law of total expectation:

$$E[FP] = N\pi_0 \int_{x \geq \gamma} f_0(x) dx, \quad E[FN] = N\pi_1 \int_{x < \gamma} f_1(x) dx$$

- $E[FP]/N$ = area of the scaled class-0 bell **to the right** of γ .
- $E[FN]/N$ = area of the scaled class-1 bell **to the left** of γ .

Normal tail areas you need (one side, beyond k SDs from the mean):

Distance from mean	Mass within $\pm k$ SD	Mass beyond, one side
1 SD	68%	16%
2 SD	95.4%	2.3%
3 SD	99.7%	0.15%

Example: class 0 $\sim N(-1, \frac{1}{4})$ (SD $\frac{1}{2}$), class 1 $\sim N(1, \frac{1}{4})$, equal priors $\rightarrow \gamma = 0$, which is 2 SDs from each mean $\rightarrow FPR = FNR = 0.023$, $TPR = 0.977$, error rate = 0.023.

6.8 Choosing a threshold τ when costs differ

The Bayes classifier (binary) = predict 1 when $\Pr(Y = 1 | x) \geq \frac{1}{2}$; it is optimal only when an FP and an FN cost the same. Generalize:

$$C_\tau(x) = 1 \text{ if } \Pr(Y = 1 | X = x) \geq \tau, \text{ else } 0, \quad \tau \in [0, 1]$$

τ	Behaviour	FNR	FPR
0	Everyone predicted 1	0	1
$\frac{1}{2}$	Bayes classifier (Default: FPR 0.2%, FNR 75%)		
$\rightarrow 1$	Everyone predicted 0 (null classifier)	1	0

As τ rises, FNR rises and FPR falls. A doctor or bank that fears FNs should **lower τ** : fewer FNs, more FPs, a higher total error rate, but a lower total **cost**.

6.9 ROC and AUC

- **ROC curve:** for every τ , plot **FPR on the x-axis vs. TPR on the y-axis**.
- $(0, 0)$ = predict all 0 ($\tau = 1$); $(1, 1)$ = predict all 1 ($\tau = 0$). The Bayes classifier on Default sits near $(0.002, 0.24)$.
- **Ideal corner (0, 1):** no false positives and every positive caught.

- **AUC** = area under the ROC; a measure of the whole family of classifiers. Perfect = 1.
- **Coin-flip classifier** (predict 1 with probability τ , ignoring x): $\text{TPR} = \text{FPR} = \tau$, so the ROC is the **diagonal** and **AUC** = $\frac{1}{2}$. Any classifier that ignores x lands on that diagonal.

Quiz problems, solved

All 59 quiz questions from Modules 2–5. Each one is laid out as **Question**, then **Answer**, then **Why**. The three you missed are marked **MISSED**. Cover the answer and try it first.

Module 2 quizzes

Quiz: Regression Function

Setup (Questions 1–2): Model $Y = X^2 + \varepsilon$, where $\varepsilon \sim \text{Unif}[-2, 2]$. Two estimators: $f_1(x) = x$ and $f_2(x) = x^2$. Hint: $E[\varepsilon] = 0$ and $E[\varepsilon^2] = \text{Var}(\varepsilon) = 4/3$.

Q1. What is $E[(Y - f_1(X))^2 \mid X = x]$?

Options: $(x^2 - x)^2 + 4/3 \cdot x - x^2 + 4/3 \cdot x^2 + 4/3 \cdot (x + 4/3)^2 \cdot 4/3$

Answer: $(x^2 - x)^2 + 4/3$

Why: $E[(x^2 + \varepsilon - x)^2] = (x^2 - x)^2 + 2(x^2 - x)E[\varepsilon] + E[\varepsilon^2] = (x^2 - x)^2 + 0 + 4/3$. That is $\text{miss}^2 + \text{Var}(\varepsilon)$.

Q2. What is $E[(Y - f_2(X))^2 \mid X = x]$? (Same options.)

Answer: $4/3$

Why: $f_2 = x^2$ is the true f , so the miss is 0: $E[(x^2 + \varepsilon - x^2)^2] = E[\varepsilon^2] = 4/3$. Only the irreducible error is left.

Quiz: Curse of Dimensionality

Setup (Questions 1–5): $H_1 = [-1, 1]^p$ is the p -dimensional hypercube. Points $a = (a_1, \dots, a_p)$ and $b = (b_1, \dots, b_p)$ are drawn uniformly at random from it. $H_r = [-r, r]^p$ with $r < 1$. Hint: the probability a uniform point lands in $T \subset S$ equals $\text{Vol}(T)/\text{Vol}(S)$.

Q1. What is the probability of the event $a \notin H_r$?

Options: $r^p \cdot \pi r^p \cdot 1 - r^p \cdot r^{p-1}$

Answer: $1 - r^p$

Why: $P(a \in H_r) = \text{Vol}(H_r)/\text{Vol}(H_1) = (2r)^p/2^p = r^p$, so the complement is $1 - r^p$.

Q2. For what value of r is $P(a \notin H_r) = 90\%$, as a function of p ?

Options: $r = 10^{-(1/p)} \cdot r = 90^{-(1/p)} \cdot r = 0.1 \cdot r = 0.9$

Answer: $r = 10^{-(1/p)}$

Why: $1 - r^p = 0.9 \rightarrow r^p = 0.1 = 10^{-1} \rightarrow$ take the p -th root.

Q3. Using that r , what is the limit of r as $p \rightarrow \infty$? Think about the interpretation.

Options: $0 \cdot 1 \cdot 0.5 \cdot 1/p$

Answer: 1

Why: $10^{-(1/p)} \rightarrow 10^0 = 1$. Even a cube almost as big as the whole space leaves 90% of the points outside it: in high dimensions nearly all the mass sits in a thin layer near the surface.

Q4. Compute the expected squared distance $E[\|a - b\|^2]$. Hint: $\|a - b\|^2 = \sum (a_i - b_i)^2$, and each coordinate is uniform on $[-1, 1]$ with mean 0 and variance $1/3$.

Options: $p/3 \cdot p/2 \cdot 2p/3 \cdot p$

Answer: $2p/3$

Why: $E\|a - b\|^2 = p \cdot E[(a_1 - b_1)^2] = p(E[a_1^2] + E[b_1^2] - 2E[a_1]E[b_1]) = p(1/3 + 1/3 - 0)$. The cross term splits because a and b are independent.

Q5. Compute the conditional expectation $E[\|a - b\|^2 \mid b = (0, \dots, 0)]$. (Same options.)

Answer: $p/3$

Why: With b at the origin, $\|a - b\|^2 = \sum a_i^2$, whose mean is $p \cdot E[a_1^2] = p/3$. Distances grow linearly with p , so a random point is far from the origin and from every other point.

Quiz: Bias-Variance Tradeoff

Setup (Questions 1–3): A graph against flexibility shows three curves: **A** is U-shaped and on top, **B** starts high and falls toward 0, **C** starts near 0 and rises; a dashed horizontal line sits at 1.0. Options for each: Squared Bias · Test MSE · Variance.

Q1. Line A is...

Answer: Test MSE

Why: It has a minimum at the optimal flexibility and stays above the dashed irreducible-error line.

Q2. Line B is...

Answer: Squared Bias

Why: More flexibility lets the model track the middle of the point cloud, so bias falls.

Q3. Line C is...

Answer: Variance

Why: More flexibility makes the fit wigglier, so it changes more from one training set to another.

Module 3 quizzes

Quiz: Linear Algebra

Setup (Questions 1–3): column vectors $x = [1, 3, 5]^T$ and $y = [2, 4, 6]^T$.

Q1. What is $x^T y$, the transpose of x multiplied by y ? (Whole number.)

Answer: 44

Formula: $x^T y = \sum x_i y_i$ (multiply matching entries, then add).

Q2. What is $y^T x$? (Whole number.)

Answer: 44

Why: The inner product is symmetric: $y^T x = x^T y$.

Q3. Evaluate $\sum_{k=1}^3 x_k y_k$.

Answer: 44

Why: It is the same dot product written as a sum.

Q4. A has 3 rows and 2 columns, B has 2 rows and 5 columns, C has 5 rows and 5 columns. Which are true? (Select all.)

1. AB exists and has 3 rows and 5 columns.
2. $A + B$ does not exist because their dimensions are not equal.

3. $AB = BA$
4. $A(BC) = (BA)C$
5. $(AB)^T = B^T A^T$
6. $A^T B = B^T A$

Answer: 1, 2 and 5

Why:

Statement	Verdict	Reason
1	True	$(3 \times 2)(2 \times 5) \rightarrow 3 \times 5$
2	True	Sums need identical sizes
3	False	$BA = (2 \times 5)(3 \times 2)$: inner sizes $5 \neq 3$, doesn't exist
4	False	BA doesn't exist (and order can't be swapped anyway)
5	True	Transposing a product reverses the order
6	False	$A^T B = (2 \times 3)(2 \times 5)$ and $B^T A = (5 \times 2)(3 \times 2)$: neither exists

Q5. Evaluate $M^T M$, where M has rows $[1, 0]$, $[1, 0]$, $[1, 10]$, $[1, 10]$.

Options: $[[20, 4], [200, 20]] \cdot [[20, 4], [20, 200]] \cdot [[4, 20], [20, 200]] \cdot [[4, 20], [200, 20]]$

Answer: $[[4, 20], [20, 200]]$

Formula: entry (i, j) of $M^T M$ = (column i of M) $\cdot$ (column j of M). $M^T M$ is always symmetric.

Quiz: Linear Regression

Setup (Questions 1–4): $D_{Tr} = \{(0, 1), (0, 3), (10, 10), (10, 6)\}$, one predictor ($p = 1$); use the univariate formulas.

Q1. What is $\bar{x}$? (Whole number.)

Answer: 5

Formula: $\bar{x} = \sum x_i / N$.

Q2. What is $\bar{y}$? (Whole number.)

Answer: 5

Formula: $\bar{y} = \sum y_i / N$.

Q3. What is β_1^* ? (One decimal.)

Answer: 0.6

Formula: $\beta_1^* = \sum (x_i - \bar{x})(y_i - \bar{y}) / \sum (x_i - \bar{x})^2 = 60/100$.

Q4. What is β_0^* ? (Whole number.)

Answer: 2

Formula: $\beta_0^* = \bar{y} - \beta_1^* \bar{x}$.

Setup (Questions 5–8): now use $\hat{\beta} = (M_X^T M_X)^{-1} M_X^T y$ with $\hat{\beta} = [\beta_0^*, \beta_1^*]^T$, M_X rows $[1, x_i]$, $y = [y_1, \dots, y_4]^T$.

Q5. What is M_X^T ?

Options include $[0, 0, 10, 10]$ (one row) and $[[1, 1, 1, 1], [0, 0, 10, 10]]$.

Answer: $[[1, 1, 1, 1], [0, 0, 10, 10]]$

Why: M_X is 4×2 (a column of 1s, then the x 's); its transpose is 2×4 .

Q6. What is $M_X^T M_X$?

Options: $[[4, 20], [20, 200]] \cdot [[20, 4], [20, 200]] \cdot [[20, 4], [200, 20]] \cdot [[4, 20], [200, 20]]$

Answer: $[[4, 20], [20, 200]]$

Formula: $M_X^T M_X = [[N, \sum x], [\sum x, \sum x^2]]$.

Q7. What is $(M_X^T M_X)^{-1}$? Hint: $[[a, b], [c, d]]^{-1} = 1/(ad - bc) \cdot [[d, -b], [-c, a]]$.

Options: $[[0.5, -0.05], [-0.05, 0.01]] \cdot [[0.1, -0.05], [-0.01, 0.05]] \cdot [[1, 5], [0.1, 0.25]] \cdot [[-0.05, 0.01], [0.5, -0.05]]$

Answer: $[[0.5, -0.05], [-0.05, 0.01]]$

Formula: $[[a, b], [c, d]]^{-1} = [[d, -b], [-c, a]]/(ad - bc)$, with $\det = 400$.

Q8. What is $M_X^T y$?

Options: $[20, 160]^T \cdot [160, 20]^T \cdot [400, 200]^T \cdot [20, 200]^T$

Answer: $[20, 160]^T$

Formula: $M_X^T y = [\sum y, \sum xy]^T$.

Quiz: Coefficient Uncertainty

Q1. In $SD(\hat{\beta}_1)^2 = \text{Var}(\hat{\beta}_1 | \{x_i\}) = \sigma^2 / \sum (x_i - \bar{x})^2$ and $SD(\hat{\beta}_0)^2 = \sigma^2 (1/n + \bar{x}^2 / \sum (x_i - \bar{x})^2)$, what is the interpretation of σ^2 ?

1. The variance of the beta coefficients arising from matrix multiplication
2. The standard deviation of the beta coefficients, which follow a normal distribution
3. The variance of the measurement noise ϵ in a linear additive model
4. The standard deviation of the measurement noise ϵ in a linear additive model

Answer: 3

Why: $\sigma^2 = \text{Var}(\epsilon)$. It is a variance (squared), not a standard deviation, and it belongs to the noise, not the coefficients.

Quiz: Hypothesis Testing

Q1. A regression of sales on TV, radio and newspaper gives:

	Coefficient	Std. dev.	t-statistic
Intercept	2.939	0.3119	9.42
TV	0.046	0.0014	32.81
radio	0.189	0.0086	21.89
newspaper	-0.001	0.0059	-0.18

What conclusions can be drawn at 95% confidence? (Select all.)

- a. The intercept is significant and should be included because its coefficient is greater than 2 (reject $\beta_0 = 0$).
- b. TV is a significant predictor because the magnitude of its t-statistic is greater than 2.
- c. We fail to reject $\beta_{\text{radio}} = 0$, so radio can be removed.
- d. Under $\beta_{\text{radio}} = 0$, we have strong evidence to reject and conclude radio affects sales.
- e. The 95% CI for newspaper is roughly $[-0.0128, 0.0108]$.

Answer: b, d, e

Why:

Option	Verdict	Reason
a	False	Right conclusion, wrong reason: significance comes from $ t = 9.42 > 2$, not the coefficient's size
b	True	$ t = 32.81$
c	False	$t = 21.89 > 2$, so reject
d	True	Same t-statistic
e	True	$-0.001 \pm 2 \times 0.0059$

Setup (Questions 2–10): nature uses $Y = \beta_0 + \beta_1 X + \varepsilon$, $\varepsilon \sim N(0, \sigma^2)$. $D_{Tr} = \{(-2, -7/2), (-1, -3), (0, 0), (1, 3), (2, 7/2)\}$. Here $\bar{x} = 0$ and $\bar{y} = 0$.

Q2. What is β_1^* ? (Whole number.)

Answer: 2

Formula: $\beta_1^* = \Sigma(x - \bar{x})(y - \bar{y}) / \Sigma(x - \bar{x})^2$; with $\bar{x} = \bar{y} = 0$ this is $\Sigma xy / \Sigma x^2 = 20/10$.

Q3. What is β_0^* ? (Whole number.)

Answer: 0

Why: $\bar{y} - \beta_1^* \bar{x} = 0 - 2 \times 0$.

Q4. Compute the RSS. (One decimal.)

Answer: 2.5

Formula: $RSS = \Sigma(y_i - \hat{y}_i)^2$, with $\hat{y} = 2x$.

Q5. Compute the RSE. (One decimal.)

Answer: 0.9

Formula: $RSE = \sqrt{RSS / (N - p - 1)} = \sqrt{(2.5/3)}$.

Q6. Compute the TSS. (One decimal.)

Answer: 42.5

Formula: $TSS = \Sigma(y_i - \bar{y})^2$.

Q7. Compute R^2 . (Two decimals.)

Answer: 0.94

Formula: $R^2 = 1 - RSS/TSS$.

Setup (Questions 8–10): 95% CIs are $\beta_0 \in [\hat{\beta}_0 \pm 2 \cdot SD(\hat{\beta}_0)]$ and $\beta_1 \in [\hat{\beta}_1 \pm 2 \cdot SD(\hat{\beta}_1)]$. Use the RSE as the estimate of the SD of ε .

Q8. Compute $SD(\hat{\beta}_1)$. (One decimal, rounded.)

Answer: 0.3

Formula: $SD(\hat{\beta}_1)^2 = RSE^2 / \Sigma(x - \bar{x})^2 = (5/6)/10$.

MISSED: you answered 0.4, which is $SD(\hat{\beta}_0)$. The slope's formula is the simple one with no bracket.

Q9. Compute $SD(\hat{\beta}_0)$. (One decimal.)

Answer: 0.4

Formula: $SD(\hat{\beta}_0)^2 = RSE^2(1/N + \bar{x}^2/\Sigma(x - \bar{x})^2) = (5/6)(1/5)$.

MISSED: you answered 0.3, the slope's value. The intercept's formula has the bracket $(1/N + \dots)$.

Q10. Can you reject $H_0: \beta_1 = 0$?

Answer: Yes

Formula: $t = \hat{\beta}_1/SD(\hat{\beta}_1) \approx 6.9 > 2$, so reject.

Module 4 quizzes

Quiz: More Categorical Inputs

Q1. In linear regression, we train a model using data that is **(A)** to predict outputs that are **(B)**. Hint: qualitative variables that have been coded (class names turned into numbers) are still qualitative.

Options: (A) only quantitative, (B) only quantitative · (A) only qualitative, (B) only qualitative · (A) only quantitative, (B) quantitative or qualitative · (A) quantitative or qualitative, (B) only quantitative

Answer: (A) quantitative or qualitative, (B) only quantitative

Why: Categorical inputs enter as dummy variables, but a categorical *output* is a classification problem, not regression.

Quiz: Nonlinear Effects

Setup (Questions 1–7): fit a quadratic $\hat{Y} = \beta_0 + \beta_1 X + \beta_2 X^2$ to $D_Tr = \{(1, 2), (2, 5), (3, 13), (4, 20)\}$ using $\hat{\beta} = (M_X^T M_X)^{-1} M_X^T y$, with $\hat{\beta} = [\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2]^T$.

Q1. Step 1: what is M_X^T ?

Options: $[[1, 1, 1], [1, 2, 4], [1, 3, 9], [1, 4, 16]] \cdot [[1, 1, 1], [1, 4, 2], [1, 9, 3], [1, 16, 4]] \cdot [[1, 1, 1, 1], [1, 2, 3, 4], [1, 4, 9, 16]] \cdot [[1, 1, 1, 1], [1, 4, 9, 16], [1, 2, 3, 4]]$

Answer: $[[1, 1, 1, 1], [1, 2, 3, 4], [1, 4, 9, 16]]$

Why: M_X is 4×3 with one **row per data point**: $[1, x_i, x_i^2]$. Its transpose is 3×4 with one **row per feature**: all 1s, then the x 's, then the x^2 's (in the same order as β).

MISSED: you picked $[[1, 1, 1], [1, 2, 4], [1, 3, 9], [1, 4, 16]]$, which is M_X itself, not its transpose.

Q2. What is $M_X^T M_X$?

Options: $[[4, 30, 10], [30, 354, 100], [10, 100, 30]] \cdot [[30, 15, 16], [10, 30, 10], [1, 1, 30]] \cdot [[4, 10, 30], [10, 30, 100], [30, 100, 354]] \cdot \text{a } 4 \times 4 \text{ matrix}$

Answer: $[[4, 10, 30], [10, 30, 100], [30, 100, 354]]$

Formula: $M_X^T M_X = [[N, \Sigma x, \Sigma x^2], [\Sigma x, \Sigma x^2, \Sigma x^3], [\Sigma x^2, \Sigma x^3, \Sigma x^4]]$: sums of powers of x (here 4, 10, 30, 100, 354).

Q3. What is $(M_X^T M_X)^{-1}$? (Hint: use numpy.)

Options: $[[7.75, 1.25, -6.75], [0.25, 1.25, -6.25], [-6.75, -1.25, 1.45]] \cdot [[7.75, -6.75, 1.25], [-6.75, 6.45, -1.25], [1.25, -1.25, 0.25]] \cdot [[7.75, 1.25, -6.75], [1.25, 0.25, -1.25], [-6.75, -1.25, 6.45]] \cdot [[-6.75, -1.25, 0.25], [0.25, 1.25, -6.25], [-6.75, -1.25, 1.45]]$

Answer: $[[7.75, -6.75, 1.25], [-6.75, 6.45, -1.25], [1.25, -1.25, 0.25]]$

Why: Verified with numpy. Without a computer: the inverse of a symmetric matrix is symmetric, which leaves two options; only this one, multiplied by $M_X^T y$, reproduces $\hat{\beta}$ below.

Q4. What is $M_X^T y$?

Options: $[459, 40, 131]^T \cdot [40, 131, 459]^T \cdot [40, 459, 131]^T \cdot [131, 40, 459]^T$

Answer: $[40, 131, 459]^T$

Formula: $M_X^T y = [\sum y, \sum xy, \sum x^2 y]^T$, in the same order as the rows of M_X^T .

Q5. What is β_0 ? (One decimal.)

Answer: -0.5

Q6. What is β_1 ? (One decimal.)

Answer: 1.2

Q7. What is β_2 ? (One decimal.)

Answer: 1.0

Why (Q5–Q7): $\hat{\beta} = (M_X^T M_X)^{-1} M_X^T y = [-0.5, 1.2, 1]^T$; read off the entries in order.

Q8. What would be your prediction $\hat{Y}$ for $X = 2$?

Options: $2 \cdot 5.9 \cdot 5 \cdot 25$

Answer: 5.9

Formula: $\hat{Y} = \beta_0 + \beta_1 x + \beta_2 x^2$ at $x = 2$.

Quiz: Interaction Terms

Setup (Questions 1–4): $N = 100$ students; $X_1 = \text{GPA}$, $X_2 = \text{Gender}$ (1 female, 0 male); $Y = \text{starting salary in \$1,000s}$. Fitted model $\hat{Y} = \beta_0 + \beta_1 \text{GPA} + \beta_2 \text{Gender} + \beta_3 (\text{GPA} \times \text{Gender})$ with $\beta_0 = 35$, $\beta_1 = 5$, $\beta_2 = -15$, $\beta_3 = 5$.

Q1. For what range of GPA do females earn more (on average) than males?

Options: $X_1 > 3.0$ · $X_1 > 2.0$ · $X_1 < 3.0$ · $X_1 < 2.0$

Answer: $X_1 > 3.0$

Why: Male (Gender = 0): $\hat{Y} = 35 + 5 \cdot \text{GPA}$. Female (Gender = 1): $\hat{Y} = (35 - 15) + (5 + 5) \cdot \text{GPA} = 20 + 10 \cdot \text{GPA}$. They cross at $35 + 5g = 20 + 10g \rightarrow g = 3$; the female line is steeper, so it is higher after 3.

Q2. True or false: since β_3 for $\text{GPA} \times \text{Gender}$ is large, there is statistical evidence of a strong interaction effect.

Answer: False

Why: Significance depends on the t-statistic (β_3 / SD), not on the coefficient's size.

Q3. A new model adds a term: $\tilde{Y} = \beta_0 + \beta_1 \text{GPA} + \beta_2 \text{Gender} + \beta_3 (\text{GPA} \times \text{Gender}) + \beta_4 (\text{GPA}^2 \times \text{Gender})$. Is its training RSS $\leq$ the previous model's?

Options: Yes · No · Not enough information

Answer: Yes

Why: It contains the old model (set $\beta_4 = 0$), so it can fit the training data at least as well.

Q4. Is the new model's **test** RSS $\leq$ the previous model's? (Same options.)

Answer: Not enough information

Why: Extra flexibility may capture real curvature (lower test RSS) or overfit the training data (higher).

Module 5 quizzes

Quiz: Classification Problems

Q1. Which of the following are classification problems? (Select all.)

- a. Which factors affect students' test scores: for 1,000 students we record hours studied, hours slept, whether they ate breakfast, age, and their test score.

- b. Predict whether the S&P 500 will rise or fall tomorrow from the unemployment rate, the 4-week treasury rate, and last year's market data.
- c. Will a new product be a failure (negative profit), a moderate success (\$0–\$1M) or a great success (over \$1M)? Data: advertising spend, adoption rate, virality.
- d. Predict the amount of CO₂ in the atmosphere at year end from whether each country has eco-friendly regulation, its production, and its pollution.

Answer: b and c

Why:

Option	Type	Reason
a	Regression	Output is a numeric score
b	Classification	Output is rise/fall (binary); numeric inputs don't matter
c	Classification	Output is one of three tiers, even though profit is measured
d	Regression	Output is an amount; a binary <i>input</i> doesn't make it classification

Quiz: Bayes Classifier and Local Averaging

Setup (Questions 1–2): binary problem with $p_1(x) = \Pr(Y = 1 \mid X = x) = 1/(1 + x^2)$ and $p_0(x) = 1 - p_1(x)$. Find the Bayes classifier in the form "if $x \in [a, b]$ then $y = 1$ ".

Q1. What is a? (Integer.)

Answer: -1

Q2. What is b? (Integer.)

Answer: 1

Why (Q1–Q2): Predict 1 when $p_1 \geq p_0$, i.e. $p_1 \geq \frac{1}{2}$. $1/(1 + x^2) \geq \frac{1}{2} \rightarrow 2 \geq 1 + x^2 \rightarrow x^2 \leq 1 \rightarrow x \in [-1, 1]$.

Setup (Questions 3–6): the triangular function $T(x) = 1 + x$ on $(-1, 0]$, $1 - x$ on $(0, 1]$, 0 otherwise. Three labels {A, B, C} with $p_A(x) = T(x)$, $p_B(x) = T(x + 1)$, $p_C(x) = 1 - p_A(x) - p_B(x)$. The Bayes classifier has the form: A for $x \in (-\frac{1}{2}, \alpha]$; B for $x \in [\beta, -\frac{1}{2}]$; C for $x \leq \delta$ or $x > \varepsilon$.

How to solve: $T(x + 1)$ is T shifted **left** by 1 (peak at -1). Write each curve piece by piece and find where the top curve changes:

Interval	Curves there	Top curve switches where
$x \leq -2$	$A = 0, B = 0, C = 1$	C on top
-2 to -1	$B = x + 2$ (rising), $C = -1 - x$ (falling), $A = 0$	$x + 2 = -1 - x \rightarrow x = -3/2$
-1 to 0	$B = -x$ (falling), $A = 1 + x$ (rising), $C = 0$	$-x = 1 + x \rightarrow x = -\frac{1}{2}$
0 to 1	$A = 1 - x$ (falling), $C = x$ (rising), $B = 0$	$1 - x = x \rightarrow x = \frac{1}{2}$
$x \geq 1$	$C = 1$	C on top

Q3. What is α ? (One decimal.)

Answer: 0.5

Q4. What is β ? (One decimal.)

Answer: -1.5

Q5. What is δ ? (One decimal.)

Answer: -1.5

Q6. What is ϵ ? (One decimal.)

Answer: 0.5

Why (Q3–Q6): From the table: C(x) = A on $(-\frac{1}{2}, \frac{1}{2}]$, B on $(-\frac{3}{2}, -\frac{1}{2}]$, C for $x \leq -\frac{3}{2}$ or $x > \frac{1}{2}$.

Quiz: Logistic Regression (odds and log-odds)

Q1. Define $p(x) = \Pr(Y = 1 \mid X = x)$ and the odds $p(x)/(1 - p(x))$. Which graph shows $f(p) = p/(1 - p)$ for $p \in [0, 1]$?

Options: Plot C (starts very high near $p = 0$ and decreases) · Plot D (a hump peaking at $p = 0.5$) · Plot B (near 0 for small p , shooting up toward 50 as $p \rightarrow 1$)

Answer: Plot B

Why: $f(0) = 0$, $f(0.5) = 1$, and $f \rightarrow \infty$ as $p \rightarrow 1$ because the denominator goes to 0. It is increasing everywhere.

Q2. Using the log-odds as the output of a linear regression, $Y = \ln(p(X)/(1 - p(X))) = \beta_0 + \beta_1 X$, which sequence correctly derives $p(X)$ as a sigmoid?

Answer: The sequence in this order:

1. $p(X)/(1 - p(X)) = e^{\{\beta_0 + \beta_1 X\}}$
2. $p(X) = (1 - p(X)) \cdot e^{\{\beta_0 + \beta_1 X\}}$
3. $p(X) = e^{\{\beta_0 + \beta_1 X\}} - p(X) \cdot e^{\{\beta_0 + \beta_1 X\}}$
4. $p(X) \cdot (1 + e^{\{\beta_0 + \beta_1 X\}}) = e^{\{\beta_0 + \beta_1 X\}}$
5. $p(X) = e^{\{\beta_0 + \beta_1 X\}} / (1 + e^{\{\beta_0 + \beta_1 X\}})$

Why: Exponentiate, multiply out, collect the p terms on one side, factor, divide. The wrong options list the same lines out of order (e.g. factoring before expanding).

Quiz: Logistic Hypothesis Testing

Q1. Logistic regression on the Default dataset (US dollars):

	Coefficient	Std. dev.	z-statistic
Intercept	-10.8690	0.4923	-22.08
Balance	0.0057	0.0002	24.74
Income	0.0030	0.0082	0.37
Student [Yes]	-0.6468	0.2362	-2.74

What conclusions can be drawn at 95% confidence? (Select all.)

- a. For every \$333.333 increase in income, the log-odds increases by 1.
- b. For every half-cent increase in balance, the probability of default increases by 1%.
- c. Being a student is not statistically significant for default.
- d. Income is not statistically significant for default.
- e. The null hypothesis is that balance influences the output.

Answer: a and d

Why:

Option	Verdict	Reason
a	True	Each \$1 adds 0.0030 to the log-odds; $333.33 \times 0.0030 = 1$
b	False	Coefficients act on the log-odds, not the probability
c	False	$ z = 2.74 > 2$, so student is significant
d	True	$ z = 0.37 < 2$
e	False	H_0 says balance does not influence the output

The z-statistic plays the same role as the t-statistic in linear regression.

Quiz: Logistic Predictions

Setup (Questions 1–2): 200 table-tennis players; X_1 = age, X_2 = rating; win = 1, loss = 0. Fitted $\beta_0 = -1$, $\beta_1 = 0.01$, $\beta_2 = 0.0004$.

Q1. Estimate the probability that a 35-year-old with rating 1900 wins.

Options: 67% · 53% · 89% · 92%

Answer: 53%

Formula: $z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 = 0.11$, then $p = e^z / (1 + e^z)$.

Q2. What rating gives a 35-year-old a 50% chance of winning? (Whole number, four digits.)

Answer: 1,625 (the key accepts 1,600)

Formula: $50\% \Leftrightarrow \log\text{-odds } z = 0$; solve $-1 + 0.01(35) + 0.0004r = 0$ for r .

MISSED: you answered 3,399, most likely a sign slip when moving terms (using $1 + 0.35$ instead of $1 - 0.35$ gives about 3,375).

Selected module practice problems

Only problems that add something beyond the quizzes and midterm set. Skipped: Python/code exercises, open-ended application lists, plain matrix multiplication, and repeats.

Module 2 practice

P1. $Y = X^2 + \varepsilon$, $\varepsilon \sim \text{Unif}[-1, 1]$.

1. Distribution of Y given $X = \frac{1}{2} \rightarrow \text{Unif}[-\frac{3}{4}, \frac{5}{4}]$. $Y = \frac{1}{4} + \varepsilon$, so shift $[-1, 1]$ by $\frac{1}{4}$.
2. Regression function and conditional variance $\rightarrow \mathbf{f(x) = x^2}$, $\mathbf{Var(Y \mid X = x) = 1/3}$. ($Y \mid X = x) \sim \text{Unif}[x^2 - 1, x^2 + 1]$: mean = midpoint; variance = $2^2/12 = 1/3$, the same for every x .
3. Estimator $\hat{Y} = X^3$: $E[(Y - \hat{Y})^2 \mid X = x] \rightarrow \mathbf{(x^2 - x^3)^2 + 1/3}$ (miss² + $\text{Var}(\varepsilon)$).
4. Reducible error $\rightarrow \mathbf{(x^2 - x^3)^2}$; irreducible = $1/3$. The estimator X^2 makes the reducible part 0, the best possible.

P2. Flexible vs. inflexible: answers are in the table in Module 2 notes (§2.7). One-line reasons: big $N \rightarrow$ flexible OK (low variance); big p and small $N \rightarrow$ flexible overfits; highly nonlinear truth $\rightarrow$ rigid model is biased; huge $\sigma^2 \rightarrow$ flexible fits noise.

P4. Sketch five curves vs. flexibility: squared bias (down), variance (up), irreducible error (flat), test error (U, above the flat line), training error (always down, can drop below the flat line).

Module 3–4 practice

P2. Why does the least-squares line pass through $(\bar{x}, \bar{y})$? $\rightarrow \hat{y}(\bar{x}) = \beta_0 + \beta_1 \bar{x} = (\bar{y} - \beta_1 \bar{x}) + \beta_1 \bar{x} = \bar{y}$. (The key's " $\beta_1 \rightarrow 0$ " wording is wrong; the $\beta_1 \bar{x}$ terms cancel.)

P3 (quick drill). $D = \{(-1, -2), (0, 0), (1, 5)\}$. $\bar{x} = 0, \bar{y} = 1 \rightarrow \beta_1 = (2 + 0 + 5)/2 = \mathbf{3.5}$, $\beta_0 = \mathbf{1}$; $\hat{y} = -2.5, 1, 4.5 \rightarrow \text{RSS} = 0.25 + 1 + 0.25 = \mathbf{1.5}$; $\text{RSE} = \sqrt{(1.5/1)} = \mathbf{1.22}$; $\text{TSS} = 9 + 1 + 16 = \mathbf{26}$; $R^2 = 24.5/26 = \mathbf{0.94}$.

P4. $N = 100$, one input; fit linear and cubic.

Truth	Compare	Answer	Reason
Linear	Training RSS	Cubic lower	More flexible always fits training data at least as well
Linear	Test RSS	Linear lower (expected)	Cubic overfits a linear truth
Unknown, nonlinear	Training RSS	Cubic lower	Same as above, whatever the truth
Unknown, nonlinear	Test RSS	Not enough information	Depends on how far from linear the truth is

P5. Test $H_1: \beta_1 = 1$ and $H_2: \beta_1 = 2$ with one CI = $\beta_1 \pm 2\text{SE}$.

$\hat{\beta}_1, \text{SE}$	95% CI	$H_1 (\beta_1 = 1)$	$H_2 (\beta_1 = 2)$
-3, 1	[-5, -1]	Reject	Reject
$\frac{1}{2}, 2$	[-3.5, 4.5]	Fail to reject	Fail to reject
2, $\frac{1}{2}$	[1, 3]	Fail to reject (1 is on the edge)	Fail to reject

Failing to reject both at once shows why "fail to reject" never means "accept": both can't be true.

P6. Fit $\hat{Y} = \alpha X^2 + \gamma$ to $(0, 0), (1, \frac{1}{2}), (-1, \frac{3}{2})$. Substitute $Z = X^2 \rightarrow$ points $(0, 0), (1, \frac{1}{2}), (1, \frac{3}{2})$. $\bar{z} = 2/3, \bar{y} = 2/3$; $z - \bar{z} = -2/3, 1/3, 1/3$; $y - \bar{y} = -2/3, -1/6, 5/6$. Σ products = $4/9 - 1/18 + 5/18 = 2/3$; $\Sigma(z - \bar{z})^2 = 6/9 = 2/3 \rightarrow \alpha = \mathbf{1}$, $\gamma = 2/3 - 2/3 = \mathbf{0}$.

Module 5–6 practice

P1. $R(x) = 1$ on $[-2, 2]$. $R(x + 1) = 1$ on $[-3, 1]$: a **left** shift (plug $x = -3$: $R(-2) = 1$).

P2. Sigmoid identities: $S(x) = 1/(1 + e^{-x}) = e^x/(e^x + 1)$; $S(-x) = 1/(1 + e^x) = e^{-x}/(e^{-x} + 1)$; $\mathbf{1 - S(x) = S(-x)}$ (common denominator: $(1 + e^{-x} - 1)/(1 + e^{-x})$).

P3. $P(Y = +1) = P(Y = -1) = \frac{1}{2}$; $X | Y = +1 \sim N(3, 1)$; $X | Y = -1 \sim N(-3, 1)$.

- Joint: $P(X = x, Y = y) = P(x | y)P(y) = (1/(2\sqrt{2\pi})) \cdot e^{-\{(x - 3y)^2/2\}}$ for $y \in \{-1, +1\}$.
- Bayes rule: equal priors and variances $\rightarrow$ threshold at the midpoint $\mathbf{x = 0}$: predict +1 for $x > 0$, -1 for $x < 0$.
- Mass of $N(-3, 1)$ on the wrong side ($x > 0$): 0 is 3 SDs away $\rightarrow (1 - 0.997)/2 \approx \mathbf{0.0015}$.

P5. Logistic: $\beta_0 = -6$, $\beta_1 = 0.05$ (hours), $\beta_2 = 1$ (GPA).

- 40 hours, GPA 3.5 $\rightarrow z = -6 + 2 + 3.5 = -0.5 \rightarrow p = e^{-0.5}/(1 + e^{-0.5}) \approx \mathbf{0.378}$.
- Hours for 50% $\rightarrow$ log-odds = $\ln(0.5/0.5) = 0 = -6 + 0.05h + 3.5 \rightarrow \mathbf{h = 50}$.

P7. Channel: $X = 2Y + \varepsilon$, $Y \in \{0, 1\}$ equally likely, $\varepsilon \sim N(-1, \frac{1}{4})$ (SD $\frac{1}{2}$).

- $Y = 0 \rightarrow X \sim N(-1, \frac{1}{4})$; $Y = 1 \rightarrow X \sim N(1, \frac{1}{4})$. So $f_0(x) = \sqrt{(2/\pi)} \cdot e^{-2(x + 1)^2}$, $f_1(x) = \sqrt{(2/\pi)} \cdot e^{-2(x - 1)^2}$; $\pi_0 = \pi_1 = \frac{1}{2}$. (The noise mean -1 shifts both class means.)
- LDA: $\hat{Y} = \arg\max_k \pi_k f_k(x)$. Equal priors and variances $\rightarrow$ midpoint: $\mathbf{\hat{Y} = 1 \text{ if } x \geq 0, \text{ else } 0}$.

3. 0 is 2 SDs ($2 \times \frac{1}{2} = 1$) from each mean $\rightarrow$ one-sided tail 0.023. **FPR = 0.023, TPR = 0.977** (FNR = 0.023 by symmetry).
4. Error rate $\rightarrow \frac{1}{2}(0.023) + \frac{1}{2}(0.023) = \mathbf{0.023}$.

Midterm practice problems, fully worked

The closest preview of the exam. Do each one on paper before reading the solution.

Problem 1

1. Predict Philadelphia house prices from bedrooms, bathrooms, school scores for 1,000 houses $\rightarrow$ **Regression** (numeric output), **n = 1000, p = 3**. The goal "which factors affect prices" is inference.
2. $p_1(x) = (x^2 - \frac{1}{2})/(x^2 + \frac{1}{2})$, defined for $|x| \geq 1/\sqrt{2}$. Bayes: predict 1 when $p_1 \geq \frac{1}{2}$.
 - $(x^2 - \frac{1}{2})/(x^2 + \frac{1}{2}) \geq \frac{1}{2} \rightarrow 2x^2 - 1 \geq x^2 + \frac{1}{2} \rightarrow x^2 \geq 3/2$.
 - **C(x) = 1 for $x \geq \sqrt{6}/2$ or $x \leq -\sqrt{6}/2$ ($\approx \pm 1.22$); 0 otherwise.**
3. Medical: FNs cost much more than FPs $\rightarrow$ **True**. A missed illness goes untreated; a false alarm only means more tests.

Problem 2 (linear regression with unknown noise level s)

$Y = \beta_0 + \beta_1 X + \varepsilon$, $\varepsilon \sim N(0, 8s^2)$. $D = \{(-1, -3), (0, -2), (1, 0), (2, 2), (3, 3)\}$.

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	product	$(x_i - \bar{x})^2$	$\hat{y}_i$	residual	$(y_i - \bar{y})^2$
-1	-3	-2	-3	6	4	-3.2	0.2	9
0	-2	-1	-2	2	1	-1.6	-0.4	4
1	0	0	0	0	0	0	0	0
2	2	1	2	2	1	1.6	0.4	4
3	3	2	3	6	4	3.2	-0.2	9
Sum				16	10			26

1. $\bar{x} = 1, \bar{y} = 0 \rightarrow \hat{\beta}_1 = 16/10 = 1.6, \hat{\beta}_0 = 0 - 1.6 \times 1 = -1.6$.
2. **RSS** = $0.04 + 0.16 + 0 + 0.16 + 0.04 = \mathbf{0.4}$; **RSE** = $\sqrt{(0.4/3)} = \sqrt{(2/15)} \approx \mathbf{0.365}$; **R²** = $1 - 0.4/26 = \mathbf{64/65} \approx \mathbf{0.985}$.
3. The noise variance is **given** ($\sigma^2 = 8s^2$), so use it, not RSE²:
 - $SD(\hat{\beta}_1) = \sqrt{(8s^2/10)} = \mathbf{(2\sqrt{5}/5)s \approx 0.894s}$
 - $SD(\hat{\beta}_0) = \sqrt{(8s^2(1/5 + 1^2/10))} = \sqrt{(2.4s^2)} = \mathbf{(2\sqrt{15}/5)s \approx 1.549s}$
 - CIs: **$\beta_1 \in 1.6 \pm 2(0.894s)$, $\beta_0 \in -1.6 \pm 2(1.549s)$.**
4. Reject $H_0: \beta_1 = 4 \Leftrightarrow 4$ outside the CI. 4 is above 1.6, so need $4 > 1.6 + 1.789s \rightarrow \mathbf{s < 2.4/1.789 = 3\sqrt{5}/5 \approx 1.34}$.
5. Cannot reject $H_0: \beta_0 = 2 \Leftrightarrow 2$ inside the CI: $2 \leq -1.6 + 3.098s \rightarrow \mathbf{s \geq 3.6/3.098 = 3\sqrt{15}/10 \approx 1.16}$.

Problem 3 (classifier on 13 points)

Red = $Y = 1$, blue = $Y = 0$. $g_1(x) = \Pr(Y = 0 \mid x) = \sigma(2 + x_1 - x_2)$.

1. **Boundary**. $\Pr(Y = 0) \geq \frac{1}{2} \Leftrightarrow 2 + x_1 - x_2 \geq 0 \Leftrightarrow x_2 \leq x_1 + 2$. So the line **$x_2 = x_1 + 2$** (through $(-2, 0)$ and $(0, 2)$); **predict 1 above it, 0 on/below it**. (Defining g_1 as $\Pr(Y = 0)$ is consistent with this, not a typo.)

2. Points (approximate coordinates read from the figure) and their predictions:

Point	Label	$x_1 + 2$ vs. x_2	Predicted	Outcome
(-3, 2)	1	$-1 < 2 \rightarrow$ above	1	TP
(-2, 2)	1	$0 < 2 \rightarrow$ above	1	TP
(-2, 1)	1	$0 < 1 \rightarrow$ above	1	TP
(-0.5, 3)	1	$1.5 < 3 \rightarrow$ above	1	TP
(-1, 0)	1	$1 > 0 \rightarrow$ below	0	FN
(1, 1)	1	below	0	FN
(2, 1)	1	below	0	FN
(-2, 3)	0	$0 < 3 \rightarrow$ above	1	FP
(0, 3)	0	$2 < 3 \rightarrow$ above	1	FP
(0, 1), (1.7, 2), (3, 1.7), (3, 0)	0	below	0	TN $\times$ 4

FN = 3, FP = 2; confusion table (predicted rows, actual columns): Predicted 1: TP 4, FP 2; Predicted 0: FN 3, TN 4.

3. Flipped classifier $C_{NB} = 1 - C_B$: every TP becomes FN, $FN \rightarrow TP$, $FP \rightarrow TN$, $TN \rightarrow FP$. **FN = 4, FP = 4**;

Predicted 1: TP 3, FP 4; Predicted 0: FN 4, TN 2.

4. Always predict 1: no negatives predicted. **FNR = 0/7 = 0**; **FPR = 6/6 = 1** (the (1, 1) corner of the ROC).

Problem 4 (short answers)

- a. $p_1 = 1/(1 + x^2) \geq 1/2 \Leftrightarrow x^2 \leq 1 \rightarrow \mathbf{C(x) = 1}$ on $[-1, 1]$, **0 elsewhere**.
- b. Spam: FPR = share of real emails flagged as spam; FNR = share of spam let through; TPR = share of spam caught.
- c. Hot chocolate sales and gas spending both rise in winter: strongly correlated, neither causes the other (season is the common cause). **Correlation does not imply causation**.
- d. Independent $\Rightarrow$ always uncorrelated. Dependent does **not** imply correlated: $X_2 = X_1^2$ with X_1 symmetric around 0 is fully dependent but has zero correlation (correlation measures only linear relationships).
- e. $Y = X + X^2 + \varepsilon$, $\varepsilon \sim N(1, 2) \rightarrow \mathbf{f(x) = x + x^2 + 1}$; irreducible error = **Var(ε) = 2**.
- f. Fit models across flexibility levels, compute test MSE for each, and pick the flexibility that **minimizes test MSE**.
- g. "A very flexible model overfits because bias is high" $\rightarrow$ **False**. Overfitting = high variance; high bias = underfitting.
- h. BOC: assign the label with the highest conditional class probability. It minimizes FN + FP combined (the error rate), but **not** FNs alone: always predicting positive gives zero FNs.

Problem 5 (likelihood of a biased coin)

- a. H, H, T (independent flips) $\rightarrow \mathbf{p^2(1 - p)}$.
- b. 20 heads and 10 tails in any order $\rightarrow \mathbf{C(30, 20) \cdot p^{20}(1 - p)^{10}}$; $C(30, 20)$ counts the orderings. (Both are maximized at $\hat{p} = 2/3$, the observed fraction of heads.)

Problem 6 (interaction with gender)

$\hat{Y} = 35 + 5 \cdot \text{GPA} - 15 \cdot \text{Gender} + 5 \cdot (\text{GPA} \times \text{Gender})$, Gender = 1 female.

- **a.** $\hat{Y}_{\text{female}} = 20 + 10 \cdot \text{GPA}$; $\hat{Y}_{\text{male}} = 35 + 5 \cdot \text{GPA}$. On $[0, 4]$, the female line starts lower (20 vs. 35) and is steeper.
- **b.** Males earn more when $35 + 5g > 20 + 10g \rightarrow \text{GPA} < 3.0$.
- **c.** Adding $\beta_4(\text{GPA}^2 \times \text{Gender})$: training RSS $\leq$ the old one (more flexible).
- **d.** Male-only data: every Gender term is 0, so both models reduce to $\beta_0 + \beta_1 \cdot \text{GPA} \rightarrow$ **same training RSS**.
- **e.** With **test** RSS the answers change: the new model may overfit (higher) or fit a truly nonlinear truth better (lower); can't tell.

Problem 7 (linear regression, noise $N(0, w^2)$)

$D = \{(-2, -3.5), (-1, -3), (0, 0), (1, 3), (2, 3.5)\}$ (same data as the Hypothesis Testing quiz).

- **a.** $\bar{x} = \bar{y} = 0$, $\sum x_i^2 = 10$, $\sum x_i y_i = 20 \rightarrow \beta_1^* = 2$, $\beta_0^* = 0$.
- **b.** $\sigma^2 = w^2$ (given): $\text{SD}(\beta_1^*) = w/\sqrt{10}$; $\text{SD}(\beta_0^*) = w\sqrt{(1/5 + 0)} = w/\sqrt{5}$. $\beta_1 \in [2 - 2w/\sqrt{10}, 2 + 2w/\sqrt{10}]$; $\beta_0 \in [-2w/\sqrt{5}, 2w/\sqrt{5}]$.
- **c.** Reject $\beta_1 = 0 \leftrightarrow 0$ below the lower end: $0 < 2 - 2w/\sqrt{10} \rightarrow w < \sqrt{10} \approx 3.16$.
- **d.** Noise $N(1, w^2)$: rewrite $Y = (\beta_0 + 1) + \beta_1 X + \varepsilon'$ with $\varepsilon' \sim N(0, w^2)$. The noise mean is **absorbed into the intercept**. Slope estimate, all SDs and the β_1 CI are unchanged; only the intercept shifts by 1. The key states it as "y's shift up by 1 $\rightarrow \beta_0^* = 1$ " (new data from the shifted model); equivalently, with these data, $\beta_0^* = 0$ now estimates $\beta_0 + 1$, so the CI for the true β_0 shifts down by 1.

Problem 8 (coin-flip classifiers and ROC)

Predict 1 with probability p ("heads"), ignoring X .

- **a.** Among actual positives a fraction p are predicted 1 $\rightarrow \text{TPR} = p$. Among actual negatives also a fraction $p \rightarrow \text{FPR} = p$. So **TPR = FPR**, and the ROC is the **straight diagonal line**. (Class proportions cancel.)
- **b.** $\text{AUC} = \int_0^1 p \, dp = 1/2$, regardless of class balance.
- **c. Corrected answer:** without using X , **no**: any classifier that ignores X makes predictions independent of the true label, so $\text{TPR} = \text{FPR}$ and it sits on the diagonal with AUC exactly $1/2$. Neither higher nor lower is possible. To get above $1/2$ you must use X (a good model); a model using X can also be flipped to get below $1/2$. The key's "increase TPR, decrease FPR" misses that this requires information about X .

Formula sheet

Read this in the last hour before the exam.

Topic	Formula / rule
Regression function	$f(x) = E[Y \mid X = x]$; noise mean is absorbed into f
Expected squared error	$E[(Y - g(x))^2 \mid X = x] = (f(x) - g(x))^2 + \text{Var}(\epsilon)$
Uniform $[a, b]$	mean $(a + b)/2$; variance $(b - a)^2/12$
Second moment	$E[Z^2] = \text{Var}(Z) + (E[Z])^2$
Curse of dimensionality	Expected points in radius- r window = $N r^p$; $P(\text{inside } [-r, r]^p) = r^p$; $E\ a - b\ ^2 = 2p/3$ on $[-1, 1]^p$
Bias-variance	Test MSE = $\text{Var}(\hat{f}) + \text{Bias}^2 + \text{Var}(\epsilon)$; overfit = high variance, underfit = high bias
Slope / intercept	$\hat{\beta}_1 = \Sigma(x - \bar{x})(y - \bar{y})/\Sigma(x - \bar{x})^2$; $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$
Matrix form	$\hat{\beta} = (M_X^T M_X)^{-1} M_X^T y$; M_X has a column of 1s; one row per point
2×2 inverse	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} / (ad - bc)$
Fit quality	$\text{RSS} = \Sigma(y - \hat{y})^2$; $\text{TSS} = \Sigma(y - \bar{y})^2$; $R^2 = 1 - \text{RSS}/\text{TSS}$; $\text{RSE} = \sqrt{\text{RSS}/(N - p - 1)}$
SD of slope	$\text{SD}(\hat{\beta}_1)^2 = \sigma^2 / \Sigma(x - \bar{x})^2$
SD of intercept	$\text{SD}(\hat{\beta}_0)^2 = \sigma^2(1/N + \bar{x}^2 / \Sigma(x - \bar{x})^2)$
Which σ^2	Given noise variance if stated, else RSE^2
95% CI	$\hat{\beta} \pm 2 \cdot \text{SD}(\hat{\beta})$
Test	t (or z) = $\hat{\beta}/\text{SD}$; $ t > 2 \rightarrow \text{reject } H_0: \beta = 0$; or null value outside CI
Solve for s	Reject $\rightarrow s < \dots$; cannot reject $\rightarrow s > \dots$
K-level category	$K - 1$ dummies; coefficients = differences from the baseline
Interaction with dummy D	Slope for $D = 1$ is $\beta_x + \beta_{\text{interaction}}$; intercept $\beta_0 + \beta_D$
Hierarchy principle	Keep main effects whenever their interaction is in the model
Bayes classifier	$\text{argmax}_k p_k(x)$; binary: predict 1 iff $p_1(x) \geq 1/2$
Logistic	$p_1 = e^z / (1 + e^z)$, $z = \beta_0 + \beta_1 x_1 + \dots$; log-odds $\ln(p/(1 - p)) = z$
Logistic coefficients	+1 unit $x \rightarrow \text{log-odds} + \beta$; odds $\times e^\beta$ (not probability $+\beta$); 50% $\leftrightarrow z = 0$; centre at $x = -\beta_0/\beta_1$
Sigmoid	$1 - S(x) = S(-x)$
Likelihood	$L = \prod \{y=1\} p_1(x_i) \prod \{y=0\} p_0(x_i)$; coin: $p^{\text{heads}} (1 - p)^{\text{tails}}$, max at the observed fraction
LDA	$C(x) = \text{argmax}_k \pi_k f_k(x)$; $\delta_k(x) = x\mu_k/\sigma^2 - \mu_k^2/(2\sigma^2) + \ln \pi_k$
LDA threshold (2 classes)	$\gamma = (\mu_1 + \mu_2)/2 + \sigma^2 \ln(\pi_1/\pi_2)/(\mu_2 - \mu_1)$; equal priors $\rightarrow$ midpoint
LDA estimates	$\pi_k^* = N_k/N$; $\mu_k^* = \text{class mean}$; $\sigma^2 = \Sigma_k \Sigma(x - \mu_k^*)^2 / (N - K)$
Rates	$\text{TPR} = \text{TP}/(\text{TP} + \text{FN})$; $\text{FNR} = 1 - \text{TPR}$; $\text{TNR} = \text{TN}/(\text{TN} + \text{FP})$; $\text{FPR} = 1 - \text{TNR}$; error = $(\text{FP} + \text{FN})/N$

Topic	Formula / rule
Normal tails (one side)	Beyond 1 SD 16%; 2 SD 2.3%; 3 SD 0.15%
LDA expected errors	$E[FP] = N\pi_0 \int \{x \geq \gamma\} f_0$; $E[FN] = N\pi_1 \int \{x < \gamma\} f_1$
Threshold τ	Predict 1 if $\Pr(Y = 1 \mid x) \geq \tau$; lower $\tau \rightarrow$ fewer FN, more FP
ROC / AUC	$x = \text{FPR}$, $y = \text{TPR}$; ignoring X $\rightarrow$ diagonal, $\text{AUC} = \frac{1}{2}$; perfect $\text{AUC} = 1$